EPSR-test-ex6 Accumulator reset (iinit 1)

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises `iinit 1` of the `epsr` engine [1,2]: the running averages of g(r), S(Q) and energies are set to zero and the r and Q grids recalculated, while the empirical potential is kept. Users do this after changing the system or the data so that old averages do not contaminate the new ones [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `inita` (accumulators zeroed, `.inpa` not read), `get_wts` (called with `iinit`: inverse weights recomputed), `runepsr17` (`.erg` history not read, `iinit` reset after the run)
- `epsr_standalone_rev3.f`: `lsparabola1` (skipped with `iinit 1`), `norma`
- `read_and_save_files.f`: `save_files_others`

Rust (port, rust/epsr-core/src/):

- `structure.rs`: `State::inita`, `reset_accumulators`
- `epsr.rs`: `Epsr::setup` (`erg_fresh`), `Epsr::iterate`
- `weights.rs`: `get_wts`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. Physically nothing changes: configuration, reference potential and EP are those of EPSR-test; only the averages of g(r), S(Q), energy and pressure start again and the r and Q grids are recomputed from the input.

Inputs

Copy of EPSR-test with only `iinit 1` (instead of 0) in `200DMSO400Water.EPSR.inp`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Three chained passes; after pass 1 the Fortran writes `iinit 0`, so passes 2-3 test the continuation. Evidence in the Fortran reference: the `.erg` file is rewritten from the start in pass 1 and has 3 lines after pass 3 (EPSR-test: 940), while `ereq` stays at 10 and the EP coefficients are kept and refined. All 27 output files are identical after every pass.

Notes

Fortran behaviour reproduced: with `iinit 1` the `.erg` unit is reopened and overwritten rather than appended.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.